

# MUHAMMAD HISYAM

## Full Stack & AI Software Engineer — Project Portfolio Showcase

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Portfolio Web: [mhisyam.com](https://mhisyam.com) · Location: Bekasi, Indonesia

## OVERVIEW & ARCHITECTURAL PROFILE

Full Stack & AI Software Engineer and Universitas Gunadarma Computer Science graduate specializing in modern web applications, desktop cross-platform software, high-performance backends, and medical deep learning. Maintainer of 26+ open-source repositories spanning TypeScript, Rust, Python, Golang, C#/.NET, and Java.

## CORE TECHNICAL MATRIX

|                    |                                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------|
| Languages          | C#, TypeScript, JavaScript, Rust, Python, Golang, Java, PHP, PL/pgSQL, HTML5, CSS3, LaTeX                   |
| Desktop & Frontend | Tauri (Rust Runtime), Astro 5, React.js, SolidJS, SvelteKit, Next.js, Vue.js, Tailwind CSS, TanStack Router |
| Backend & Cloud    | .NET, Supabase, Node.js, Express.js, Go (Gin/Fiber), Java Spring Boot, Laravel, REST APIs, WebSockets       |
| AI & Deep Learning | PyTorch, U-Net Semantic Segmentation, OpenCV, Model Context Protocol (MCP), Google Gemini API               |
| Database & DevOps  | PostgreSQL, Supabase RLS, MongoDB, Redis, Docker, Git, GitHub Actions, Vite, Bun, Biome, Linux CLI          |

## FEATURED PROJECT SHOWCASE

### 1. RetailHub – Retail Management & POS Platform

**Tech Stack:** Tauri (Rust backend runtime), TypeScript, Vite, TanStack Router, Supabase (PostgreSQL), Tailwind CSS

**Repository:** [github.com/caleheinz25/retailhub](https://github.com/caleheinz25/retailhub)

- Engineered a high-performance cross-platform desktop Point of Sale (POS) and inventory management platform for retail operations.
- Designed Supabase PostgreSQL architecture utilizing Row Level Security (RLS) policies and custom PL/pgSQL database functions.
- Integrated real-time database subscription listeners for instant stock synchronization and transactional receipt tracking across devices.

### 2. Developer Portfolio & Interactive Dev Suite

**Tech Stack:** Astro 5, SolidJS, Tailwind CSS 4, TypeScript, HTML5 Canvas

**Repository:** [github.com/caleheinz25/portofolio-2](https://github.com/caleheinz25/portofolio-2) · **Live Web:** [mhisyam.com](https://mhisyam.com)

- Developed a ultra-fast static-first developer portfolio website using Astro 5 island architecture and SolidJS reactive components.
- Embedded an interactive Unix-style terminal shell (`hisyam-cli`), REST API request client demo, audio music player, and 8 playable mini-games (2048, Whack-a-Bug, Sudoku, Pong, Minesweeper, Snake, Memory Match, Typing Speed).
- Built automated build workflows fetching live GitHub repository statistics and profile metadata.

### 3. Pneumonia U-Net Semantic Segmentation Pipeline

**Tech Stack:** Python, PyTorch, Torchvision, OpenCV, Matplotlib, NumPy

**Repository:** [github.com/caleheinz25/pneumonia-segmentation-unet](https://github.com/caleheinz25/pneumonia-segmentation-unet)

- Built an end-to-end deep learning medical image segmentation pipeline utilizing PyTorch and U-Net encoder-decoder architecture.

- Processed chest X-ray image datasets to generate accurate binary segmentation masks highlighting pulmonary infection regions.
- Implemented evaluation metrics including Dice Similarity Coefficient (DSC) and Intersection over Union (IoU).

#### 4. Gemini Model Context Protocol (MCP) Client

**Tech Stack:** TypeScript, Node.js, Google Gemini API, MCP Protocol

**Repository:** [github.com/caleheinzz25/mcp-client-gemini](https://github.com/caleheinzz25/mcp-client-gemini)

- Implemented an extensible Model Context Protocol (MCP) client connecting agentic AI models with local tool definitions.
- Integrated Google Gemini API for structured function calling, prompt execution, and automated workflow orchestration.

#### 5. Real-Time Energy Telemetry Monitoring System

**Tech Stack:** TypeScript, React, Node.js, WebSockets, Chart.js

**Repository:** [github.com/caleheinzz25/realtime-energy-monitoring](https://github.com/caleheinzz25/realtime-energy-monitoring)

- Developed a real-time IoT energy consumption telemetry dashboard rendering high-frequency sensor streams.
- Implemented WebSocket connection handlers with automatic reconnects and dynamic threshold alert metrics.

#### 6. Golang & SvelteKit Real-Time Concurrent Chat System

**Tech Stack:** Go (Golang), Goroutines, Gorilla WebSockets, SvelteKit, TypeScript

**Repository:** [github.com/caleheinzz25/golang-sveltekit-realtime-chatapp](https://github.com/caleheinzz25/golang-sveltekit-realtime-chatapp)

- Built a concurrent multi-room real-time messaging application leveraging Golang channels and goroutines.
- Paired with a SvelteKit reactive frontend delivering instant message broadcasting and online presence indicators.

#### 7. Secure Java JWT Authentication Microservice

**Tech Stack:** Java, Spring Boot, Spring Security, JWT, PostgreSQL

**Repository:** [github.com/caleheinzz25/jwt-auth-with-refreshToken-and-versionToken](https://github.com/caleheinzz25/jwt-auth-with-refreshToken-and-versionToken)

- Designed an enterprise-grade Java security microservice implementing access tokens, sliding session refresh tokens, and instant token version invalidation.

## EDUCATION & ACADEMIC BACKGROUND

**Universitas Gunadarma**

*Bekasi, Indonesia*

*Bachelor of Computer Science (Informatika) — Fresh Graduate*

- Specialized in Software Engineering, Web Systems, and Applied Artificial Intelligence.